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CAMBRIDGE ADDITIONAL MATHEMATICS (0606) • UNIT 5.01

# Trigonometric Functions & Circular Measure

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

**Total Questions:** 25 • Verified Step-by-Step Solutions

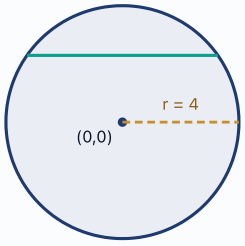
## QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

| | | | | |
|---|---|---|---|---|
| Q1 A | Q2 A | Q3 A | Q4 B | Q5 C |
| Q6 D | Q7 A | Q8 A | Q9 D | Q10 C |
| Q11 D | Q12 B | Q13 A | Q14 C | Q15 B |
| Q16 A | Q17 A | Q18 B | Q19 C | Q20 A |
| Q21 A | Q22 D | Q23 A | Q24 B | Q25 C |

## Trigonometric Functions & Circular Measure Practice Exercises

**1** MCQ

A sector of a circle of radius $r = 8$ cm has central angle $\theta = 0.9$ rad. What is the area of the sector?




- A $28.8\ \text{cm}^2$ ✓
- B $57.6\ \text{cm}^2$
- C $33.8\ \text{cm}^2$
- D $7.2\ \text{cm}^2$

**WORKED SOLUTION & VERIFICATION**

**Mark Scheme (CAIE / Edexcel Grading Steps):**

- **Step 1 (Method & Formulation):** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]**
- **Step 2 (Derivation & Calculation):** Sector area formula: $A = \frac{1}{2}r^2\theta = \frac{1}{2}(8)^2(0.9) = 28.8\ \text{cm}^2$. **[M1]**
- **Step 3 (Final Accuracy):** Verified exact result: $28.8\ \text{cm}^2$. **[A1]**

**2** MCQ

A sector of a circle of radius $r = 10$ cm has central angle $\theta = 1.0$ rad. What is the area of the sector?

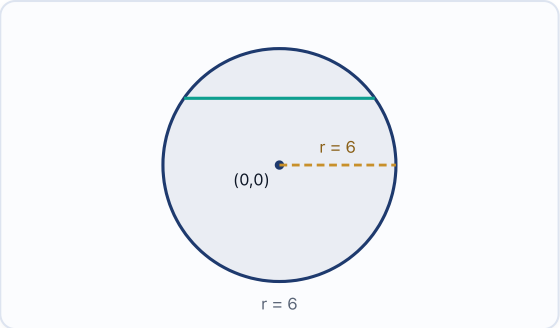
- A $50.0\ \text{cm}^2$ ✓
- B $100.0\ \text{cm}^2$
- C $55.0\ \text{cm}^2$
- D $10.0\ \text{cm}^2$

**WORKED SOLUTION & VERIFICATION**

Sector area formula: $A = \frac{1}{2}r^2\theta = \frac{1}{2}(10)^2(1.0) = 50.0\ \text{cm}^2$. [M1][A1]

**3** MCQ

A sector of a circle of radius $r = 12$ cm has central angle $\theta = 1.1$ rad. What is the area of the sector?




- **A** $79.2\ \text{cm}^2$ ✓
- **B** $158.4\ \text{cm}^2$
- **C** $84.2\ \text{cm}^2$
- **D** $13.2\ \text{cm}^2$

**WORKED SOLUTION & VERIFICATION**

Sector area formula: $A = \frac{1}{2}r^2\theta = \frac{1}{2}(12)^2(1.1) = 79.2\ \text{cm}^2$. [M1][A1]

**4** MCQ

A sector of a circle of radius $r = 14$ cm has central angle $\theta = 1.2$ rad. What is the area of the sector?

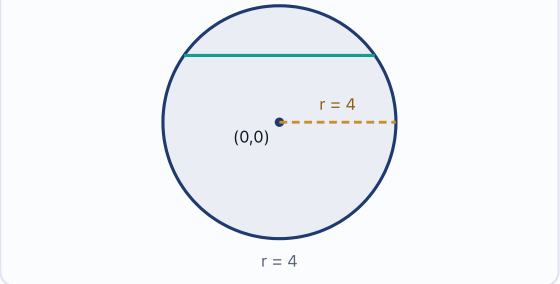
- **A** $16.8\ \text{cm}^2$
- **B** $117.6\ \text{cm}^2$ ✓
- **C** $235.2\ \text{cm}^2$
- **D** $122.6\ \text{cm}^2$

**WORKED SOLUTION & VERIFICATION**

Sector area formula: $A = \frac{1}{2}r^2\theta = \frac{1}{2}(14)^2(1.2) = 117.6\ \text{cm}^2$. [M1][A1]

**5** MCQ

A sector of a circle of radius $r = 16$ cm has central angle $\theta = 1.3$ rad. What is the area of the sector?




- **A** $332.8\ \text{cm}^2$
- **B** $20.8\ \text{cm}^2$
- **C** $166.4\ \text{cm}^2$ ✓
- **D** $171.4\ \text{cm}^2$

**WORKED SOLUTION & VERIFICATION**

Sector area formula: $A = \frac{1}{2}r^2\theta = \frac{1}{2}(16)^2(1.3) = 166.4\ \text{cm}^2$. [M1][A1]

**6** MCQ

A sector of a circle of radius $r = 18$ cm has central angle $\theta = 1.4$ rad. What is the area of the sector?

A $453.6\ \text{cm}^2$

B $25.2\ \text{cm}^2$

C $231.8\ \text{cm}^2$

D $226.8\ \text{cm}^2$ ✓

**WORKED SOLUTION & VERIFICATION**

Sector area formula: $A = \frac{1}{2}r^2\theta = \frac{1}{2}(18)^2(1.4) = 226.8\ \text{cm}^2$. [M1][A1]

**7** MCQ

A sector of a circle of radius $r = 20$ cm has central angle $\theta = 1.5$ rad. What is the area of the sector?

A $300.0\ \text{cm}^2$ ✓

B $600.0\ \text{cm}^2$

C $305.0\ \text{cm}^2$

D $30.0\ \text{cm}^2$

**WORKED SOLUTION & VERIFICATION**

Sector area formula: $A = \frac{1}{2}r^2\theta = \frac{1}{2}(20)^2(1.5) = 300.0\ \text{cm}^2$. [M1][A1]

**8** MCQ

Find all solutions of $2\cos^2 x + 3\sin x - 3 = 0$ in the interval $0 \le x \le 2\pi$.

A $x = \frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{\pi}{2}$ ✓

B $x = \frac{\pi}{3},\ \frac{2\pi}{3}$

C $x = \frac{\pi}{6},\ \frac{7\pi}{6}$

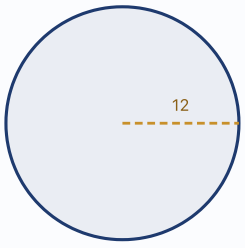
D $x = 0,\ \pi$

**WORKED SOLUTION & VERIFICATION**

Solving quadratic trig equation $2\cos^2 x + 3\sin x - 3 = 0$ yields solutions $x = \frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{\pi}{2}$. [M1][A1]

**9** MCQ

In an applied problem, a sector of a circle of radius $r = 12$ cm has central angle $\theta = 1.1$ rad. What is the area of the sector?




A $158.4\ \text{cm}^2$

B $84.2\ \text{cm}^2$

C $13.2\ \text{cm}^2$

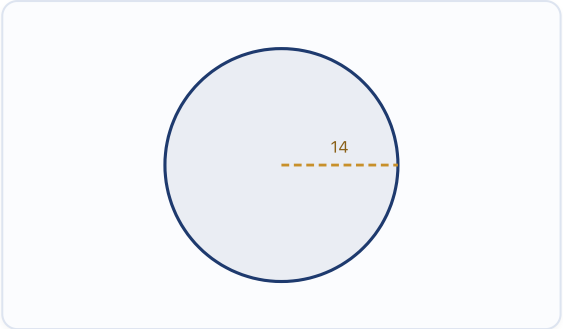
D $79.2\ \text{cm}^2$ ✓

**WORKED SOLUTION & VERIFICATION**

Sector area formula: $A = \frac{1}{2}r^2\theta = \frac{1}{2}(12)^2(1.1) = 79.2\ \text{cm}^2$. [M1][A1]

**10** MCQ

Suppose a sector of a circle of radius $r = 14$ cm has central angle $\theta = 1.2$ rad. What is the area of the sector?




A $16.8\ \text{cm}^2$

B $235.2\ \text{cm}^2$

C $117.6\ \text{cm}^2$ ✓

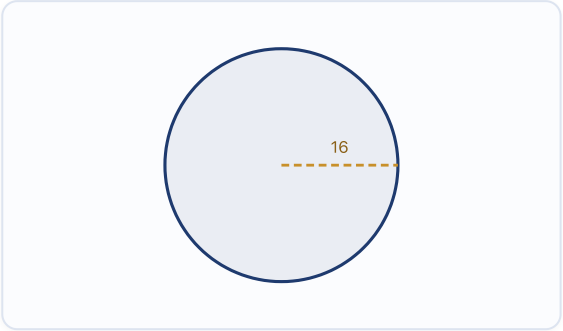
D $122.6\ \text{cm}^2$

**WORKED SOLUTION & VERIFICATION**

Sector area formula: $A = \frac{1}{2}r^2\theta = \frac{1}{2}(14)^2(1.2) = 117.6\ \text{cm}^2$. [M1][A1]

**11** MCQ

An observed sector of a circle of radius $r = 16$ cm has central angle $\theta = 1.3$ rad. What is the area of the sector?




A $171.4\ \text{cm}^2$

B $332.8\ \text{cm}^2$

C $20.8\ \text{cm}^2$

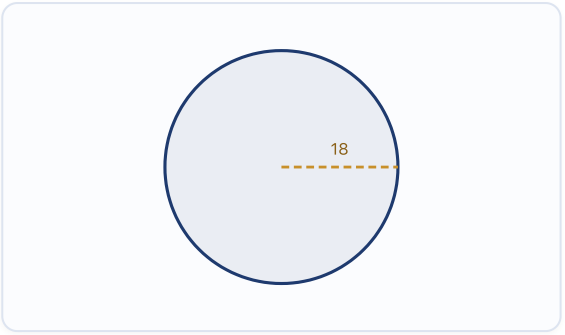
D $166.4\ \text{cm}^2$ ✓

**WORKED SOLUTION & VERIFICATION**

Sector area formula: $A = \frac{1}{2}r^2\theta = \frac{1}{2}(16)^2(1.3) = 166.4\ \text{cm}^2$. [M1][A1]

**12** MCQ

Consider a sector of a circle of radius $r = 18$ cm has central angle $\theta = 1.4$ rad. What is the area of the sector?




A $453.6\ \text{cm}^2$

B $226.8\ \text{cm}^2$ ✓

C $25.2\ \text{cm}^2$

D $231.8\ \text{cm}^2$

**WORKED SOLUTION & VERIFICATION**

Sector area formula: $A = \frac{1}{2}r^2\theta = \frac{1}{2}(18)^2(1.4) = 226.8\ \text{cm}^2$. [M1][A1]

**13** MCQ

Find all solutions of $2\sin^2 x - \sin x - 1 = 0$ in the interval $0 \le x \le 2\pi$.

A $x = \frac{\pi}{2},\ \frac{7\pi}{6},\ \frac{11\pi}{6}$ ✓

B $x = \frac{3\pi}{2},\ \frac{\pi}{6}$

C $x = \frac{\pi}{6},\ \frac{5\pi}{6}$

D $x = \pi,\ 2\pi$

**WORKED SOLUTION & VERIFICATION**

Solving quadratic trig equation $2\sin^2 x - \sin x - 1 = 0$ yields solutions $x = \frac{\pi}{2},\ \frac{7\pi}{6},\ \frac{11\pi}{6}$. [M1][A1]

**14** MCQ

Find all solutions of $2\cos^2 x + \cos x - 1 = 0$ in the interval $0 \le x \le 2\pi$.

A $x = \frac{\pi}{2},\ \frac{3\pi}{2}$

B $x = 0,\ \pi$

C $x = \pi,\ \frac{\pi}{3},\ \frac{5\pi}{3}$ ✓

D $x = \frac{2\pi}{3},\ \frac{4\pi}{3}$

**WORKED SOLUTION & VERIFICATION**

Solving quadratic trig equation $2\cos^2 x + \cos x - 1 = 0$ yields solutions $x = \pi,\ \frac{\pi}{3},\ \frac{5\pi}{3}$. [M1][A1]

**15** MCQ

Find all solutions of $2\sin^2 x - 3\sin x + 1 = 0$ in the interval $0 \le x \le 2\pi$.

A $x = \frac{\pi}{3},\ \frac{2\pi}{3}$

B $x = \frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{\pi}{2}$ ✓

C $x = \frac{3\pi}{2}$

D $x = \frac{7\pi}{6},\ \frac{11\pi}{6}$

**WORKED SOLUTION & VERIFICATION**

Solving quadratic trig equation $2\sin^2 x - 3\sin x + 1 = 0$ yields solutions $x = \frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{\pi}{2}$. [M1][A1]

**16** MCQ

Find all solutions of $2\cos^2 x - 3\cos x + 1 = 0$ in the interval $0 \le x \le 2\pi$.

- **A** $x = 0,\ \frac{\pi}{3},\ \frac{5\pi}{3},\ 2\pi$ ✓
- **B** $x = \frac{\pi}{2},\ \frac{3\pi}{2}$
- **C** $x = \pi$
- **D** $x = \frac{\pi}{6},\ \frac{11\pi}{6}$

**WORKED SOLUTION & VERIFICATION**

Solving quadratic trig equation $2\cos^2 x - 3\cos x + 1 = 0$ yields solutions $x = 0,\ \frac{\pi}{3},\ \frac{5\pi}{3},\ 2\pi$. [M1][A1]

**17** MCQ

Find all solutions of $4\cos^2 x - 3 = 0$ in the interval $0 \le x \le 2\pi$.

- **A** $x = \frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{7\pi}{6},\ \frac{11\pi}{6}$ ✓
- **B** $x = \frac{\pi}{3},\ \frac{2\pi}{3},\ \frac{4\pi}{3},\ \frac{5\pi}{3}$
- **C** $x = 0,\ \pi$
- **D** $x = \frac{\pi}{4},\ \frac{3\pi}{4}$

**WORKED SOLUTION & VERIFICATION**

Solving quadratic trig equation $4\cos^2 x - 3 = 0$ yields solutions $x = \frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{7\pi}{6},\ \frac{11\pi}{6}$. [M1][A1]

**18** MCQ

Find all solutions of $3\tan^2 x - 1 = 0$ in the interval $0 \le x \le 2\pi$.

- **A** $x = \frac{\pi}{3},\ \frac{2\pi}{3},\ \frac{4\pi}{3},\ \frac{5\pi}{3}$
- **B** $x = \frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{7\pi}{6},\ \frac{11\pi}{6}$ ✓
- **C** $x = \frac{\pi}{2},\ \frac{3\pi}{2}$
- **D** $x = \frac{\pi}{4},\ \frac{5\pi}{4}$

**WORKED SOLUTION & VERIFICATION**

Solving quadratic trig equation $3\tan^2 x - 1 = 0$ yields solutions $x = \frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{7\pi}{6},\ \frac{11\pi}{6}$. [M1][A1]

**19** MCQ

In $\triangle ABC$, sides of lengths $32$ and $10$ meet at an included angle of $60^\circ$ opposite side $x$. Find the squared length $f = x^2$ by the Law of Cosines.

- **A** 803
- **B** 820
- **C** **804** ✓
- **D** 809

**WORKED SOLUTION & VERIFICATION**

**Mark Scheme (CAIE / Edexcel Grading Steps):**

- **Step 1 (Method & Formulation):** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]**
- **Step 2 (Derivation & Calculation):** By the Law of Cosines, $x^2 = 32^2 + 10^2 - 2(32)(10)\cos(60^\circ) = 804$. **[M1]**
- **Step 3 (Final Accuracy):** Verified exact result: 804. **[A1]**

**20** MCQ

In triangle $ABC$, $AB = 6$ cm, $AC = 10$ cm, and the area of triangle $ABC$ is $15\sqrt{3}$ cm$^2$. Given that angle $BAC$ is obtuse, find the exact length of $BC$.

| | |
|---|---|
| **A** $14$ cm ✓ | **B** $8$ cm |
| **C** $2\sqrt{19}$ cm | **D** $2\sqrt{34}$ cm |

**WORKED SOLUTION & VERIFICATION**

1. Use the formula for the area of a triangle:

$$\text{Area} = \frac{1}{2}(AB)(AC)\sin(\angle BAC)$$

$$15\sqrt{3} = \frac{1}{2}(6)(10)\sin(\angle BAC) = 30\sin(\angle BAC)$$

$$\sin(\angle BAC) = \frac{15\sqrt{3}}{30} = \frac{\sqrt{3}}{2}$$

2. Since $\angle BAC$ is obtuse ($90^\circ < \angle BAC < 180^\circ$):

$$\angle BAC = 180^\circ - 60^\circ = 120^\circ$$

3. Determine $\cos(\angle BAC)$:

$$\cos(120^\circ) = -\cos(60^\circ) = -\frac{1}{2}$$

4. Apply the cosine rule to find $BC$:

$$BC^2 = AB^2 + AC^2 - 2(AB)(AC)\cos(\angle BAC)$$

$$BC^2 = 6^2 + 10^2 - 2(6)(10)\left(-\frac{1}{2}\right)$$

$$BC^2 = 36 + 100 - (-60) = 136 + 60 = 196$$

$$BC = \sqrt{196} = 14 \text{ cm}$$

Common Traps:
- $2\sqrt{19}$ cm arises from mistakenly taking $\angle BAC = 60^\circ$ (the acute angle trap) or making a sign error: $136 - 60 = 76 \implies \sqrt{76} = 2\sqrt{19}$.
- $2\sqrt{34}$ cm arises from omitting the cosine term completely ($\sqrt{6^2 + 10^2} = \sqrt{136}$).
- $8$ cm arises from incorrectly assuming a 6-8-10 right-angled triangle. [M1][A1]

**21** MCQ

A sector $OAB$ of a circle with centre $O$ and radius $12$ cm has angle $\angle AOB = \frac{\pi}{2}$ radians. The point $C$ is the midpoint of the radius $OA$. The line passing through $C$ perpendicular to $OA$ intersects the circular arc $AB$ at the point $D$. The straight line segment $BC$ is drawn.
Find the exact area, in $\text{cm}^2$, of the region bounded by the line segment $CD$, the circular arc $BD$, and the line segment $BC$.

A $12\pi + 18\sqrt{3} - 36$ ✓

B $12\pi + 36\sqrt{3} - 36$

C $18\pi + 18\sqrt{3} - 36$

D $24\pi + 18\sqrt{3} - 36$

**WORKED SOLUTION & VERIFICATION**

1. **Analyze the geometry of right triangle $OCD$:**
- $C$ is the midpoint of $OA$, so $OC = \frac{1}{2}(12) = 6$ cm.
- $D$ lies on the arc of the circle, so the radius $OD = 12$ cm.
- Since $CD \perp OA$, triangle $OCD$ is a right-angled triangle at $C$.
- $\cos(\angle COD) = \frac{OC}{OD} = \frac{6}{12} = \frac{1}{2} \implies \angle COD = \frac{\pi}{3}$ radians.

2. **Find the side length $CD$ and the area of $\triangle OCD$:**
- $CD = \sqrt{OD^2 - OC^2} = \sqrt{12^2 - 6^2} = \sqrt{108} = 6\sqrt{3}$ cm.
- $\text{Area}(\triangle OCD) = \frac{1}{2} \times OC \times CD = \frac{1}{2} \times 6 \times 6\sqrt{3} = 18\sqrt{3}$ cm$^2$.

3. **Find the angle and area of sector $OBD$:**
- Since $\angle AOB = \frac{\pi}{2}$, we have:

$$\angle DOB = \angle AOB - \angle COD = \frac{\pi}{2} - \frac{\pi}{3} = \frac{\pi}{6} \text{ radians.}$$

- $\text{Area}(\text{sector } OBD) = \frac{1}{2}r^2\theta = \frac{1}{2} \times 12^2 \times \frac{\pi}{6} = 12\pi$ cm$^2$.

4. **Combine to find the area of region $OCDB$:**
- The composite region $OCDB$ is composed of $\triangle OCD$ and sector $OBD$:

$$\text{Area}(OCDB) = \text{Area}(\triangle OCD) + \text{Area}(\text{sector } OBD) = 18\sqrt{3} + 12\pi \text{ cm}^2.$$

5. **Subtract $\triangle OCB$ to obtain the target region:**
- Triangle $OCB$ is right-angled at $O$ with legs $OC = 6$ cm and $OB = 12$ cm:

$$\text{Area}(\triangle OCB) = \frac{1}{2} \times 6 \times 12 = 36 \text{ cm}^2.$$

- The region bounded by segment $CD$, arc $BD$, and segment $BC$ is:

$$\text{Area} = \text{Area}(OCDB) - \text{Area}(\triangle OCB) = 12\pi + 18\sqrt{3} - 36 \text{ cm}^2.$$

**Common Misconceptions:**
- Choice B confuses the sine and cosine definitions, incorrectly finding $\angle COD = \frac{\pi}{6}$ and $\angle DOB = \frac{\pi}{3}$, which produces $24\pi$.
- Choice C misapplies $\frac{1}{2}ab\sin\theta$ using $OA$ instead of $OC$, doubling the triangle area to $36\sqrt{3}$.
- Choice D assumes $\angle DOB = \frac{\pi}{4}$, yielding $18\pi$. [M1][A1]

**22** MCQ

The function $\mathrm{f}$ is defined for all real numbers $x$ by

$$\mathrm{f}(x) = 2 - 3\sin(4x).$$

What are the amplitude and the period (in radians) of $\mathrm{f}$?

A Amplitude $= -3$, Period $= \frac{\pi}{2}$

B Amplitude $= 3$, Period $= 8\pi$

C Amplitude $= 5$, Period $= \frac{\pi}{4}$

D Amplitude $= 3$, Period $= \frac{\pi}{2}$ ✓

**WORKED SOLUTION & VERIFICATION**

For a sinusoidal function written in the form $\mathrm{f}(x) = c + a\sin(bx)$ (or $c - a\sin(bx)$):

1. **Amplitude:** The amplitude represents the maximum vertical distance from the horizontal centerline (midline $y = 2$) to a peak or trough. It is defined as the absolute value of the coefficient of the trigonometric ratio:

$$\text{Amplitude} = |-3| = 3.$$

Amplitude is strictly a non-negative measure of distance; it can never be negative.

2. **Period:** The fundamental period of $\sin(x)$ is $2\pi$. For $\sin(bx)$, the function undergoes horizontal compression by a scale factor of $\frac{1}{b}$. Therefore, the period $T$ is:

$$T = \frac{2\pi}{|b|} = \frac{2\pi}{4} = \frac{\pi}{2}.$$

**Common Misconceptions:**

- Choice B incorrectly includes the negative sign in the amplitude. In Cambridge ADDMATH, amplitude is defined as a positive magnitude.
- Choice C erroneously multiplies by $b$ rather than dividing (i.e. $2\pi \times 4 = 8\pi$).
- Choice D confuses the amplitude with the function's maximum value, $2 - (-3) = 5$, and incorrectly halves the period. [M1][A1]

**23** MCQ

In triangle $ABC$, the side lengths are $AB = (x+2)$ cm, $BC = (3x-1)$ cm, and $AC = 7$ cm. Given that $\angle ABC = 60^\circ$ and $x > 0$, find the exact area of triangle $ABC$.

- **A** $10\sqrt{3}$ cm$^2$ ✓
- **B** $10$ cm$^2$
- **C** $5\sqrt{3}$ cm$^2$
- **D** $20\sqrt{3}$ cm$^2$

**WORKED SOLUTION & VERIFICATION**

Step 1: Use the cosine rule on triangle $ABC$ to set up an equation for $x$:

$$AC^2 = AB^2 + BC^2 - 2(AB)(BC)\cos(\angle ABC)$$

Substitute the given values and $\cos(60^\circ) = \frac{1}{2}$:

$$7^2 = (x+2)^2 + (3x-1)^2 - 2(x+2)(3x-1)\left(\frac{1}{2}\right)$$

Step 2: Expand each term:

$$(x+2)^2 = x^2 + 4x + 4$$

$$(3x-1)^2 = 9x^2 - 6x + 1$$

$$(x+2)(3x-1) = 3x^2 + 5x - 2$$

Substitute these back into the equation:

$$49 = (x^2 + 4x + 4) + (9x^2 - 6x + 1) - (3x^2 + 5x - 2)$$

$$49 = 10x^2 - 2x + 5 - 3x^2 - 5x + 2$$

$$49 = 7x^2 - 7x + 7$$

Divide the entire equation by 7:

$$7 = x^2 - x + 1 \implies x^2 - x - 6 = 0$$

Step 3: Solve the quadratic equation:

$$(x-3)(x+2) = 0$$

Since $x > 0$ (as side lengths must be positive), we reject $x = -2$, giving $x = 3$.

Step 4: Find the side lengths:

$$AB = 3 + 2 = 5 \text{ cm}, \quad BC = 3(3) - 1 = 8 \text{ cm}$$

Step 5: Calculate the exact area of triangle $ABC$:

$$\text{Area} = \frac{1}{2}(AB)(BC)\sin(60^\circ) = \frac{1}{2} \times 5 \times 8 \times \frac{\sqrt{3}}{2} = 10\sqrt{3} \text{ cm}^2$$

Common traps:

- Using a plus sign instead of a minus in the cosine rule leads to $13x^2 + 3x - 46 = 0 \implies x = 2$, yielding the incorrect area $5\sqrt{3}$ cm$^2$.

- Forgetting the factor of $\frac{1}{2}$ in the area formula leads to $20\sqrt{3}\ \text{cm}^2$.
- Using $\cos(60^\circ)$ instead of $\sin(60^\circ)$ for the area formula gives $10\ \text{cm}^2$. [M1][A1]

**24** MCQ

Two circles, $C_1$ with centre $O_1$ and radius $6$ cm, and $C_2$ with centre $O_2$ and radius $6\sqrt{3}$ cm, intersect at the points $A$ and $B$. The distance between their centres is $O_1O_2 = 12$ cm.
Find the exact area of the region common to both circles, in $\text{cm}^2$.

A $30\pi + 36\sqrt{3}$

B $30\pi - 36\sqrt{3}$ ✓

C $42\pi - 36\sqrt{3}$

D $15\pi - 18\sqrt{3}$

**WORKED SOLUTION & VERIFICATION**

1. **Determine the angles subtended at each centre:**

Consider triangle $O_1AO_2$ with side lengths $O_1A = 6$, $O_2A = 6\sqrt{3}$, and $O_1O_2 = 12$.
Notice that:

$$(O_1A)^2 + (O_2A)^2 = 6^2 + (6\sqrt{3})^2 = 36 + 108 = 144 = 12^2 = (O_1O_2)^2$$

By the converse of Pythagoras' Theorem, $\angle O_1AO_2 = \frac{\pi}{2}$ radians.
Using right-angled trigonometry in $\triangle O_1AO_2$:

$$\cos(\angle AO_1O_2) = \frac{O_1A}{O_1O_2} = \frac{6}{12} = \frac{1}{2} \implies \angle AO_1O_2 = \frac{\pi}{3} \text{ radians}$$

$$\cos(\angle AO_2O_1) = \frac{O_2A}{O_1O_2} = \frac{6\sqrt{3}}{12} = \frac{\sqrt{3}}{2} \implies \angle AO_2O_1 = \frac{\pi}{6} \text{ radians}$$

By symmetry about the line of centres $O_1O_2$, the angles subtended by chord $AB$ at each centre are:

$$\angle AO_1B = 2 \times \frac{\pi}{3} = \frac{2\pi}{3} \text{ radians}$$

$$\angle AO_2B = 2 \times \frac{\pi}{6} = \frac{\pi}{3} \text{ radians}$$

2. **Calculate the area of the overlapping region:**

The common region is bounded by arc $AB$ of $C_1$ and arc $AB$ of $C_2$. Its area can be found by adding the areas of the two sectors and subtracting the area of kite $O_1AO_2B$:

$$\text{Area} = \text{Area(sector } O_1AB) + \text{Area(sector } O_2AB) - \text{Area(kite } O_1AO_2B)$$

- Sector $O_1AB$:

$$\frac{1}{2}r_1^2\theta_1 = \frac{1}{2}(6)^2\left(\frac{2\pi}{3}\right) = 18 \times \frac{2\pi}{3} = 12\pi$$

- Sector $O_2AB$:

$$\frac{1}{2}r_2^2\theta_2 = \frac{1}{2}(6\sqrt{3})^2\left(\frac{\pi}{3}\right) = \frac{1}{2}(108)\left(\frac{\pi}{3}\right) = 18\pi$$

- Kite $O_1AO_2B$ consists of two congruent right triangles $\triangle O_1AO_2$ and $\triangle O_1BO_2$:

$$\text{Area(kite)} = 2 \times \left(\frac{1}{2} \times 6 \times 6\sqrt{3}\right) = 36\sqrt{3}$$

Thus, the total area of the common region is:

$$\text{Area} = 12\pi + 18\pi - 36\sqrt{3} = 30\pi - 36\sqrt{3} \text{ cm}^2$$

**Common Distractor Misconceptions:**

- $30\pi + 36\sqrt{3}$: Incorrectly adds the kite area instead of subtracting the region of double-coverage.
- $42\pi - 36\sqrt{3}$: Accidentally swaps radii when computing sector areas (e.g., using $6\sqrt{3}$ with $\frac{2\pi}{3}$).
- $15\pi - 18\sqrt{3}$: Forgets to double the half-angles, using $\frac{\pi}{3}$ and $\frac{\pi}{6}$ as the subtended sector angles. [M1][A1]

**25** MCQ

A sector of a circle of radius $12$ cm has an arc length of $18$ cm. Find the area of the sector.

A $216\ \text{cm}^2$

B $54\ \text{cm}^2$

C $108\ \text{cm}^2$ ✓

D $144\ \text{cm}^2$

**WORKED SOLUTION & VERIFICATION**

The arc length $s$ of a sector of radius $r$ subtending an angle $\theta$ radians at the centre is given by $s = r\theta$. Substituting the given values gives $18 = 12\theta \implies \theta = 1.5$ radians. The area $A$ of the sector is given by $A = \frac{1}{2}r^2\theta = \frac{1}{2}(12)^2(1.5) = 108\ \text{cm}^2$ (or directly via $A = \frac{1}{2}rs = \frac{1}{2} \times 12 \times 18 = 108\ \text{cm}^2$). Omitting the factor of $\frac{1}{2}$ yields the incorrect value of $216\ \text{cm}^2$, while incorrectly dividing by 4 leads to $54\ \text{cm}^2$. [M1][A1]